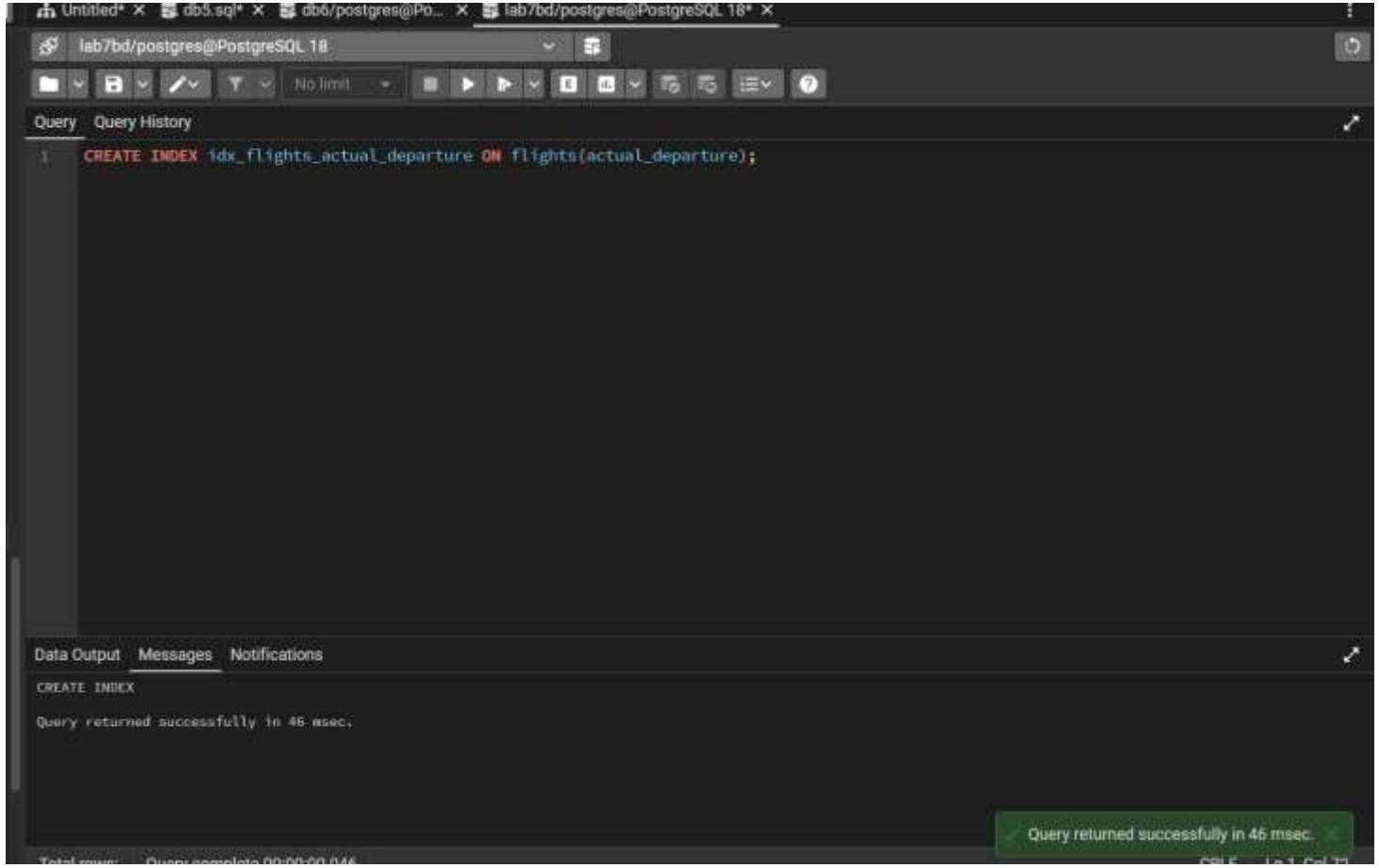
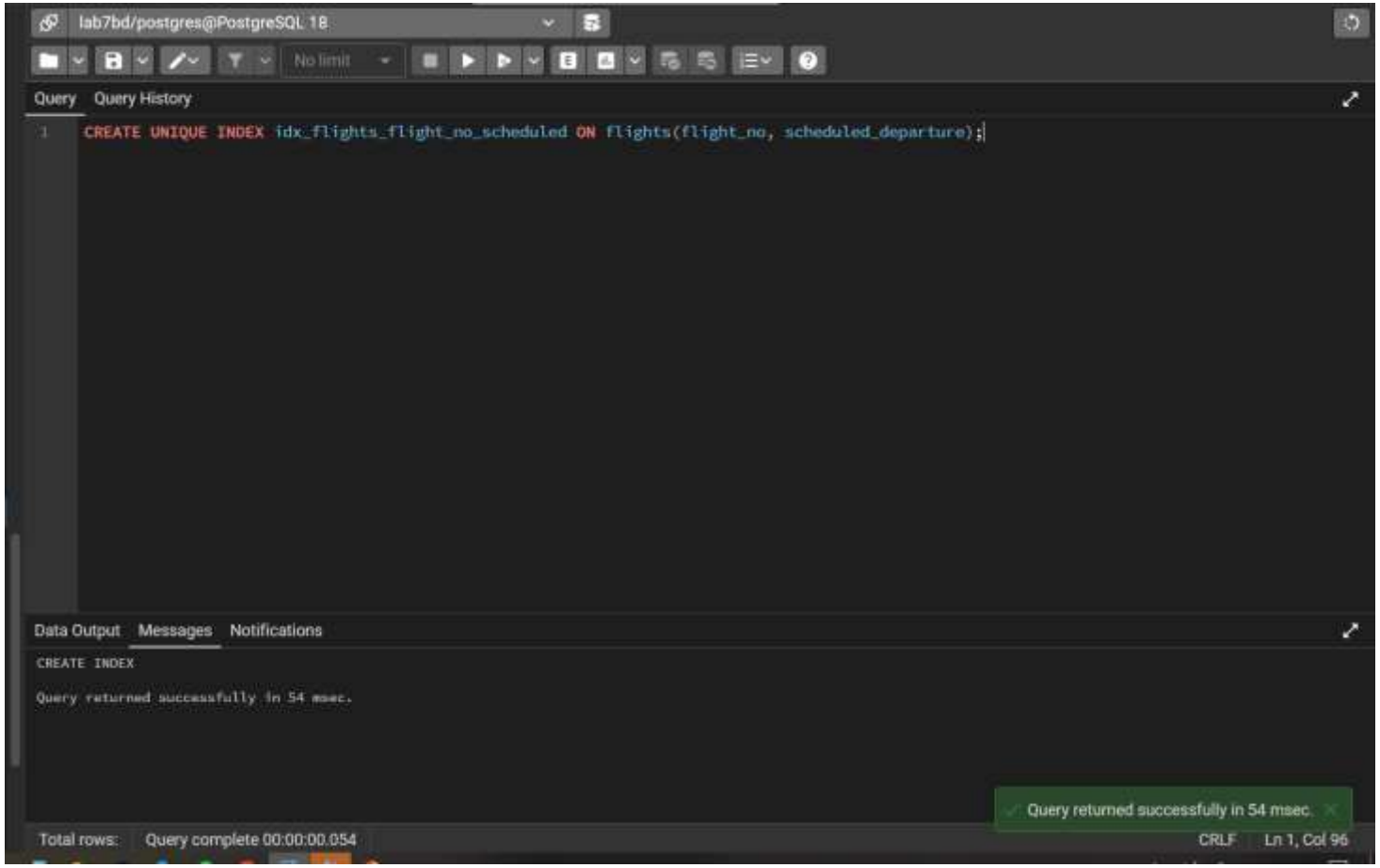
7 lab

Kairzhan Khamza

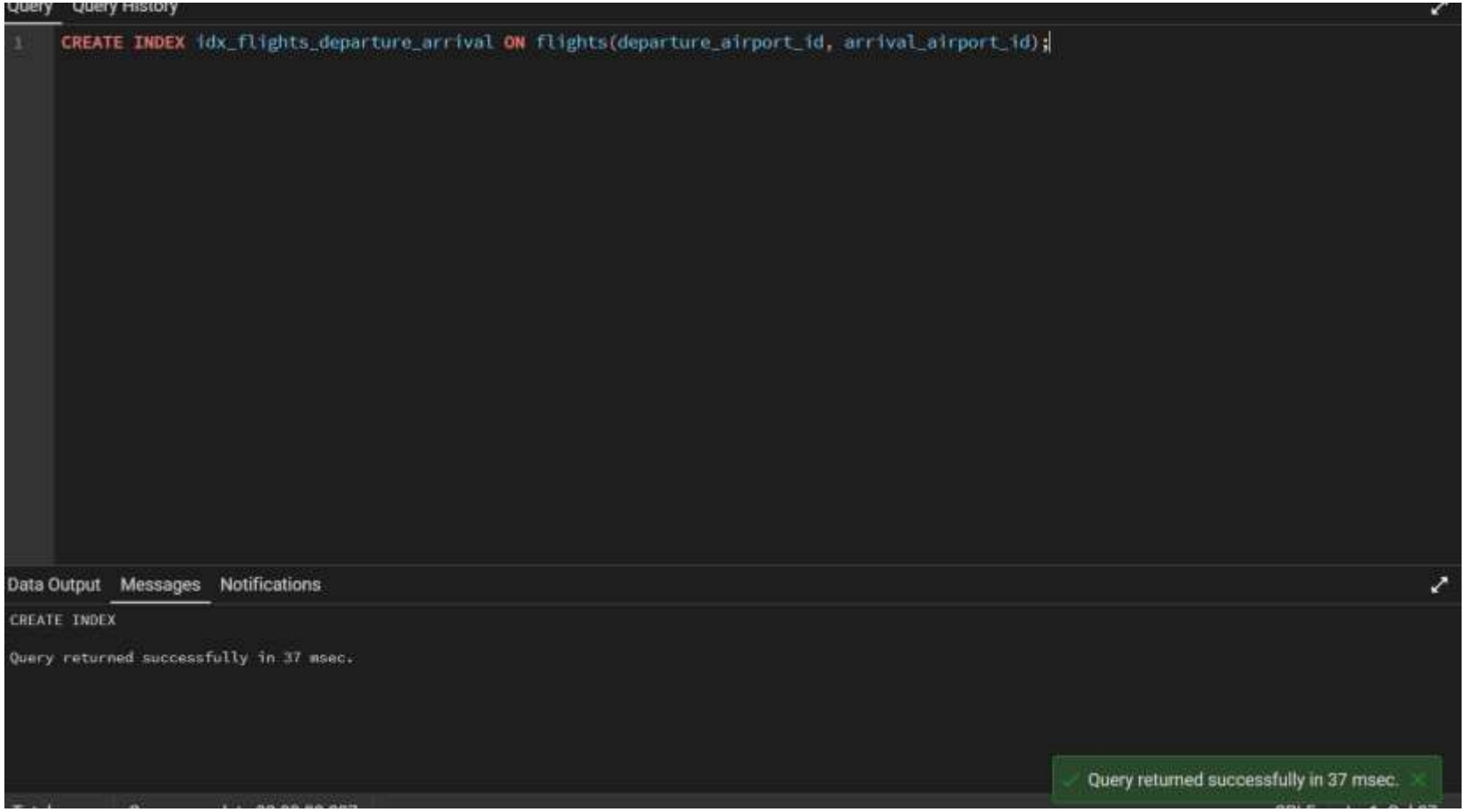
1 task


lab7bd/postgres@PostgreSQL 18
No limit
Query
Query History
1
CREATE INDEX idx_flights_actual_departure ON flights(actual_departure);
Data Output
Messages
Notifications
CREATE INDEX
Query returned successfully in 46 msec.
Query returned successfully in 46 msec.

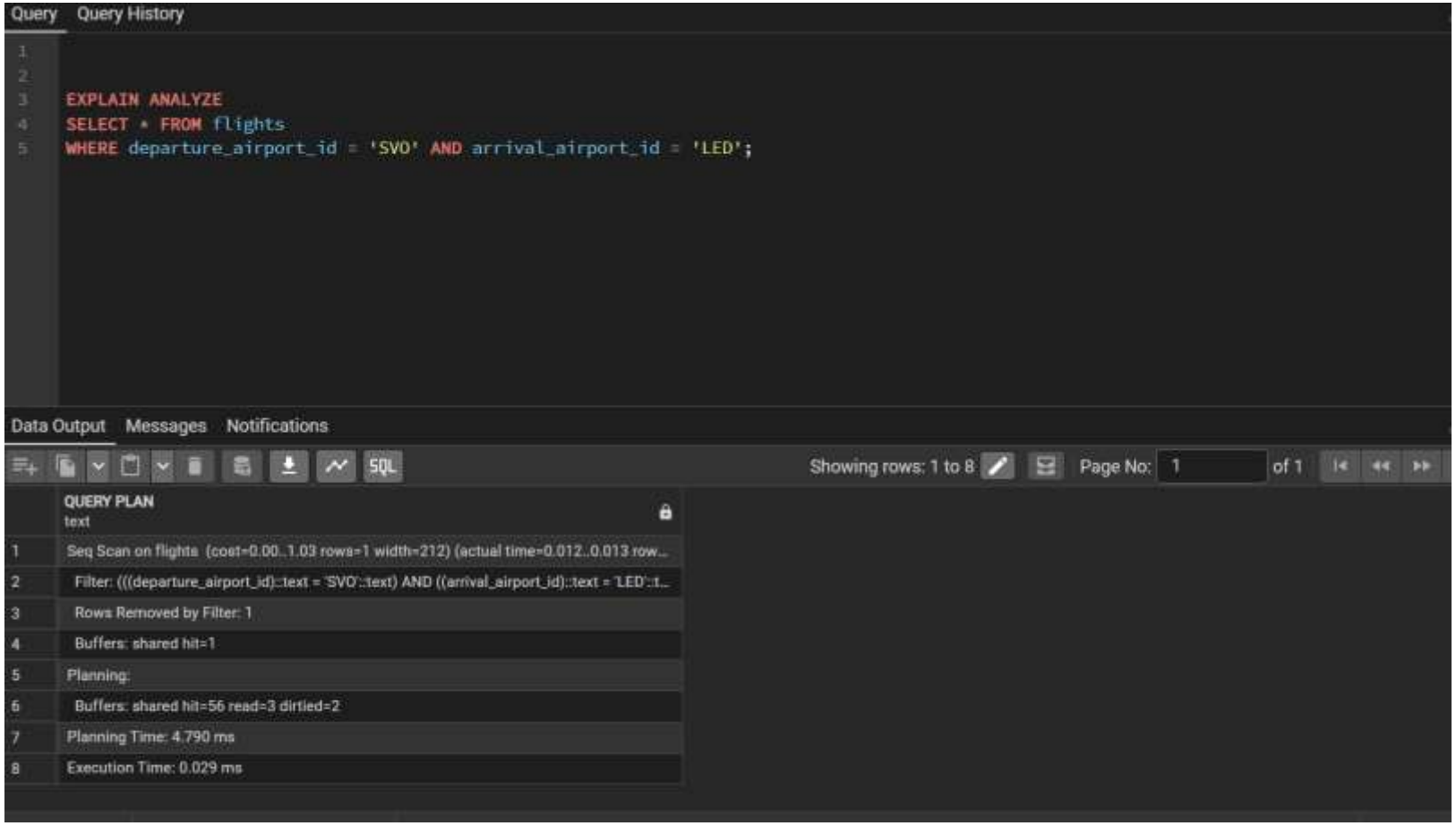
2 task


lab7bd/postgres@PostgreSQL 18
No limit
Query
Query History
1
CREATE UNIQUE INDEX idx_flights_flight_no_scheduled ON flights(flight_no, scheduled_departure);
Data Output
Messages
Notifications
CREATE INDEX
Query returned successfully in 54 msec.
Query returned successfully in 54 msec.
Total rows:
Query complete 00:00:00.054
CRLF
Ln 1, Col 96

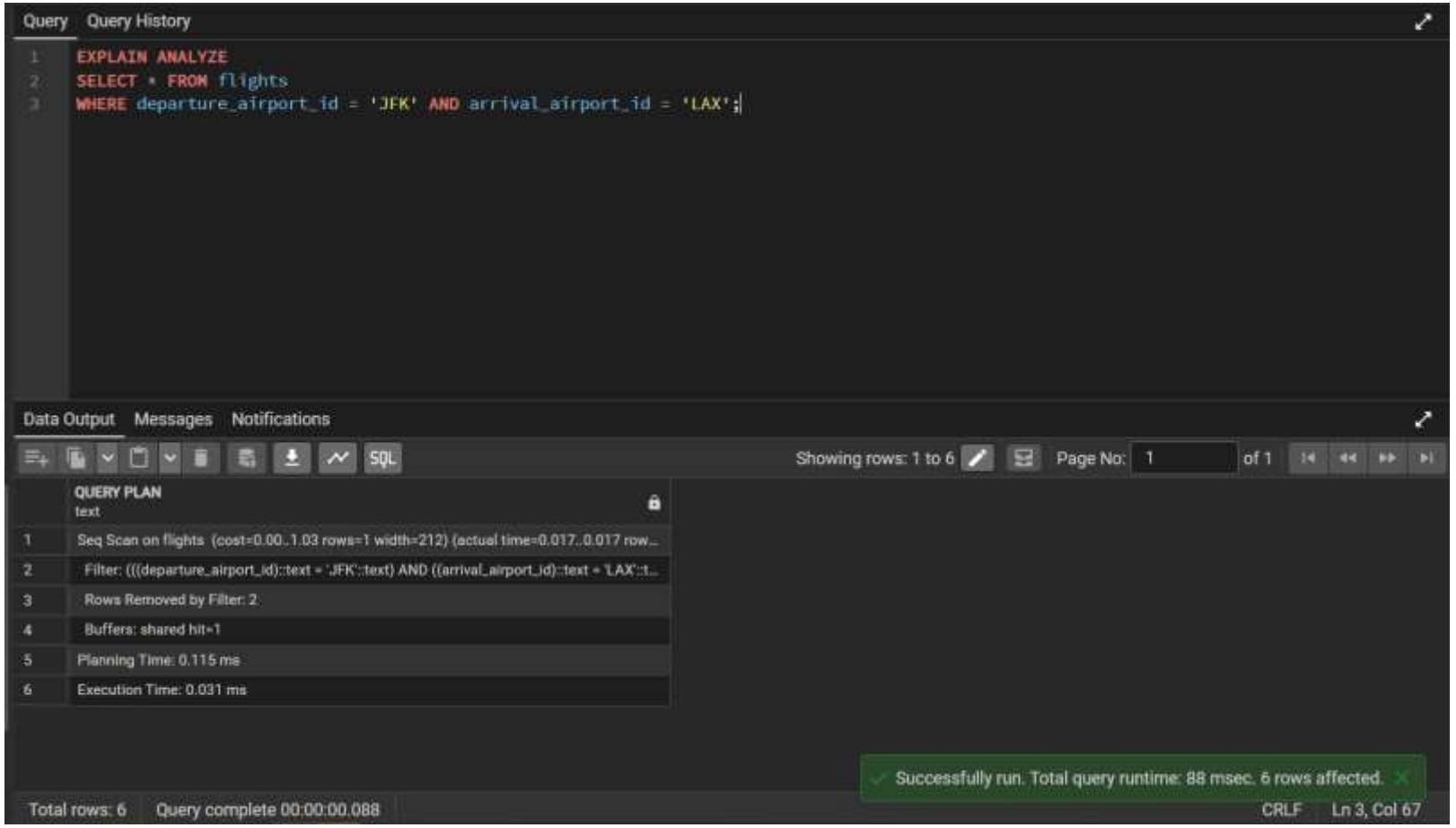
3 task


Query
Query History
1
CREATE INDEX idx_flights_departure_arrival ON flights(departure_airport_id, arrival_airport_id);
Data Output
Messages
Notifications
CREATE INDEX
Query returned successfully in 37 msec.
Query returned successfully in 37 msec.

4 task


Query
Query History
EXPLAIN ANALYZE
SELECT * FROM flights
WHERE departure_airport_id = 'SVO' AND arrival_airport_id = 'LED';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on flights (cost=0.00..1.03 rows=1 width=212) (actual time=0.012..0.013 row...
2 Filter: (((departure_airport_id)::text = 'SVO'::text) AND ((arrival_airport_id)::text = 'LED'::t...
3 Rows Removed by Filter: 1
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=56 read=3 dirtied=2
7 Planning Time: 4.790 ms
8 Execution Time: 0.029 ms

5 task


Query
Query History
1 EXPLAIN ANALYZE
2 SELECT * FROM flights
3 WHERE departure_airport_id = 'JFK' AND arrival_airport_id = 'LAX';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 6
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on flights (cost=0.00..1.03 rows=1 width=212) (actual time=0.017..0.017 row...
2 Filter: (((departure_airport_id)::text = 'JFK'::text) AND ((arrival_airport_id)::text = 'LAX'::t...
3 Rows Removed by Filter: 2
4 Buffers: shared hit=1
5 Planning Time: 0.115 ms
6 Execution Time: 0.031 ms
Successfully run. Total query runtime: 88 msec. 6 rows affected.
Total rows: 6
Query complete 00:00:00.088
CRLF
Ln 3, Col 67

6 task

Query | Query History

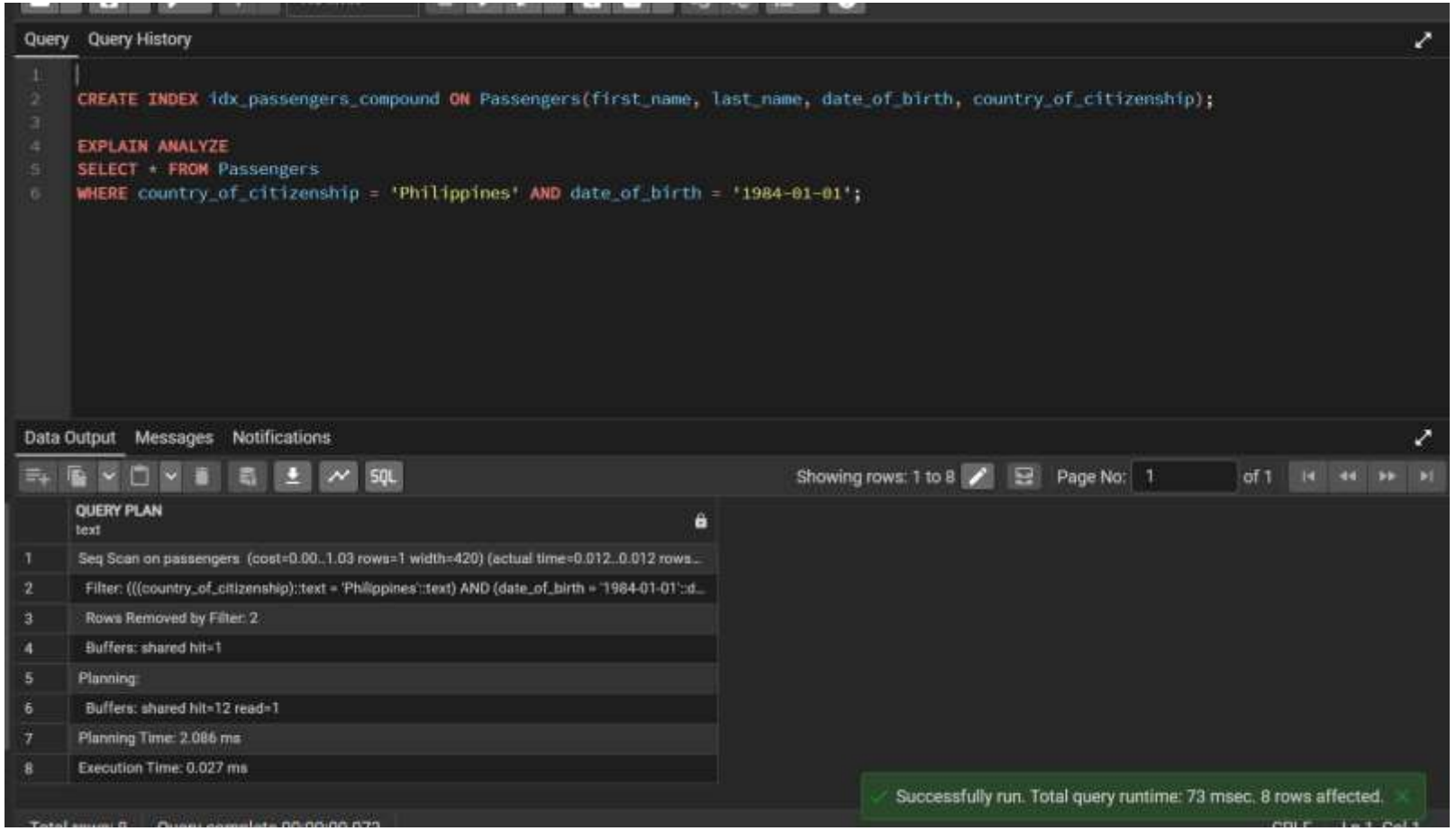
```
1
2  CREATE UNIQUE INDEX idx_passengers_passport ON Passengers(passport_number);
3
4  INSERT INTO Passengers (passenger_id, first_name, last_name, passport_number, date_of_birth, country_of_citizenship)
5  VALUES
6  (1001, 'John', 'Doe', 'AB1234567', '1990-05-12', 'USA'),
7  (1002, 'Jane', 'Smith', 'AB1234567', '1985-10-20', 'USA');
```

Data Output | Messages | Notifications

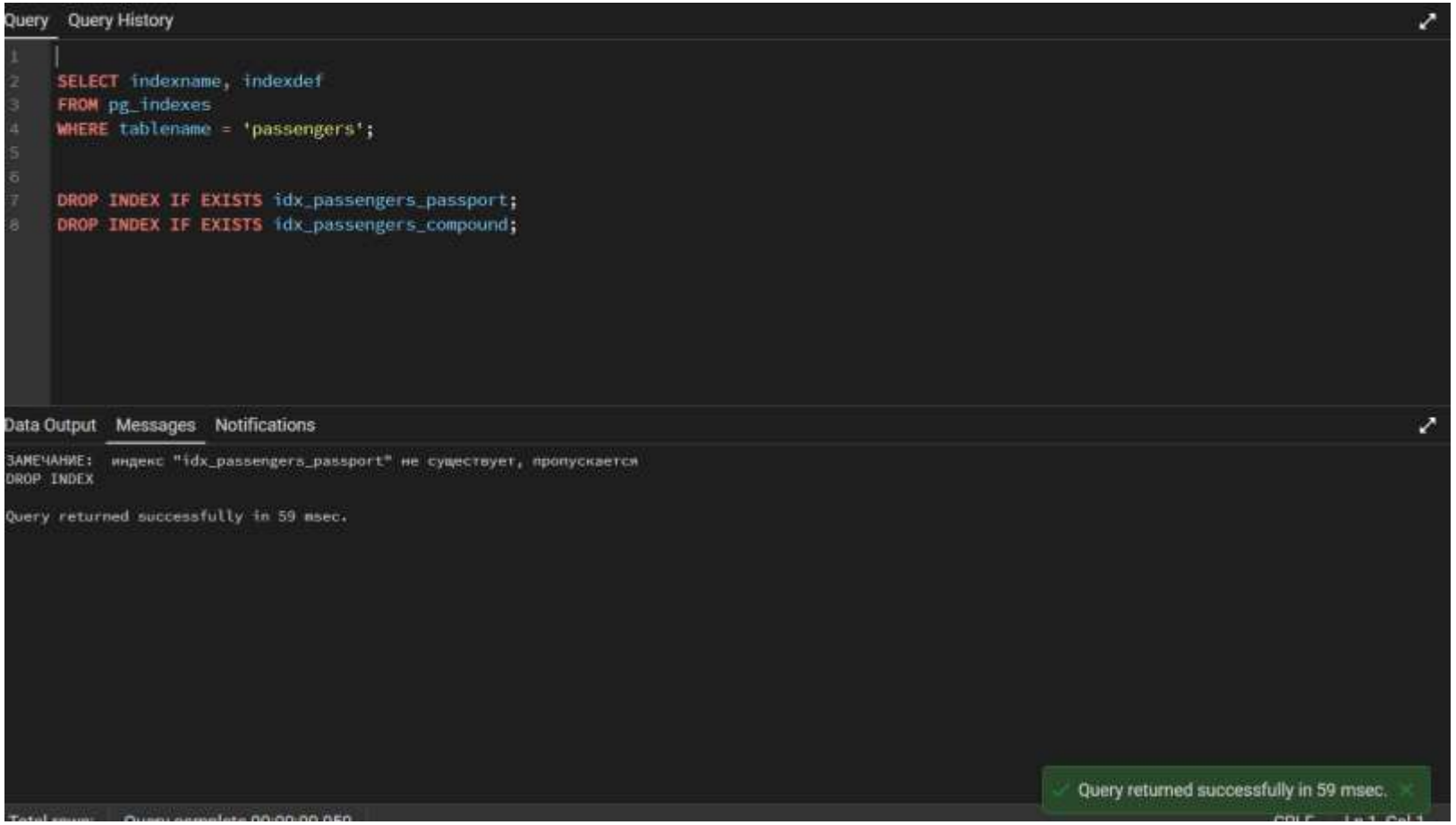
```
ERROR:  повторяющееся значение ключа нарушает ограничение уникальности "idx_passengers_passport"
Ключ "(passport_number)=(AB1234567)" уже существует.

ОШИБКА:  повторяющееся значение ключа нарушает ограничение уникальности "idx_passengers_passport"
SQL state: 23505
Detail: Ключ "(passport_number)=(AB1234567)" уже существует.
```

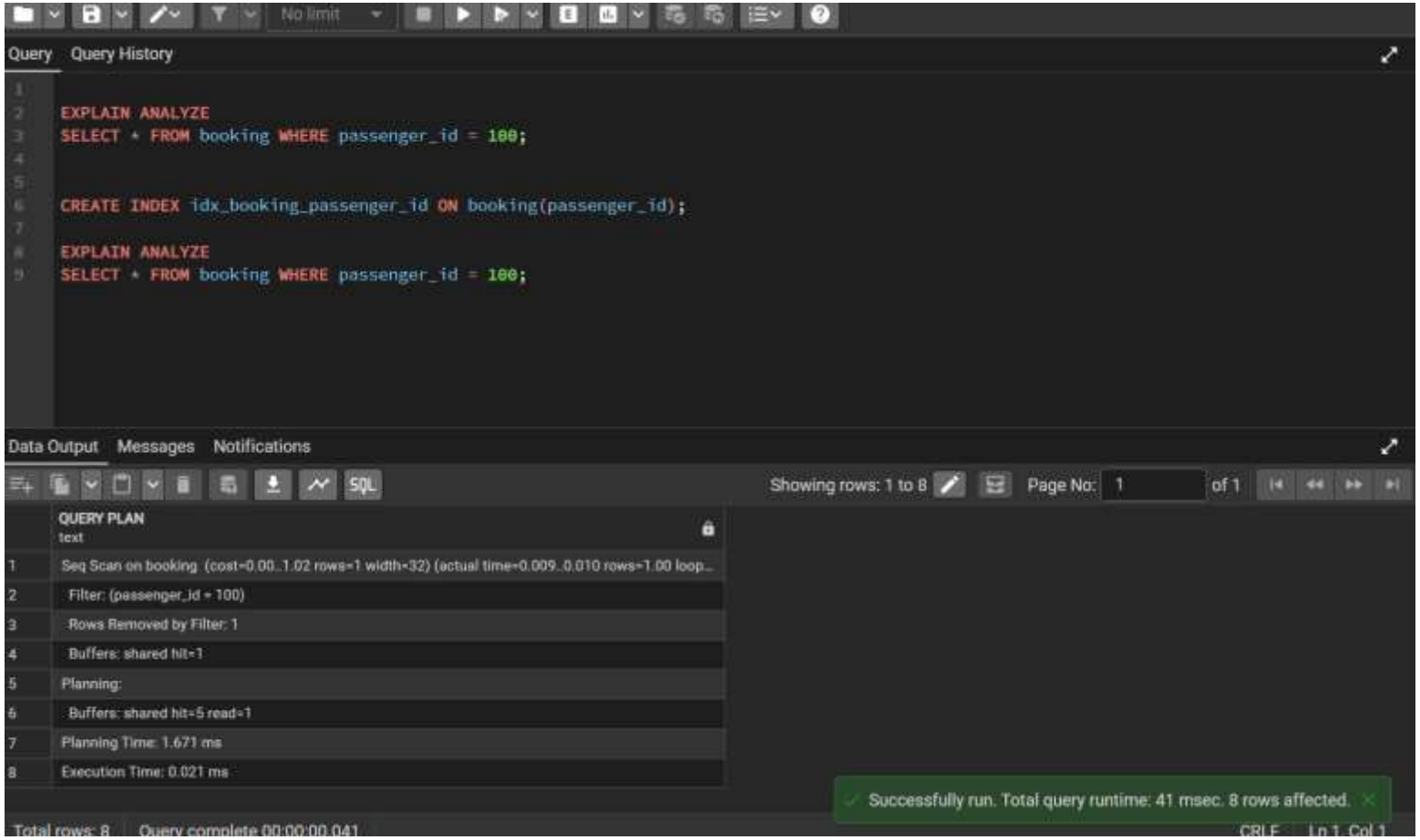
7 task


Query
Query History
CREATE INDEX idx_passengers_compound ON Passengers(first_name, last_name, date_of_birth, country_of_citizenship);
EXPLAIN ANALYZE
SELECT * FROM Passengers
WHERE country_of_citizenship = 'Philippines' AND date_of_birth = '1984-01-01';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on passengers (cost=0.00..1.03 rows=1 width=420) (actual time=0.012..0.012 rows...
2 Filter: (((country_of_citizenship)::text = 'Philippines'::text) AND (date_of_birth = '1984-01-01'::d...
3 Rows Removed by Filter: 2
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=12 read=1
7 Planning Time: 2.086 ms
8 Execution Time: 0.027 ms
Successfully run. Total query runtime: 73 msec. 8 rows affected.

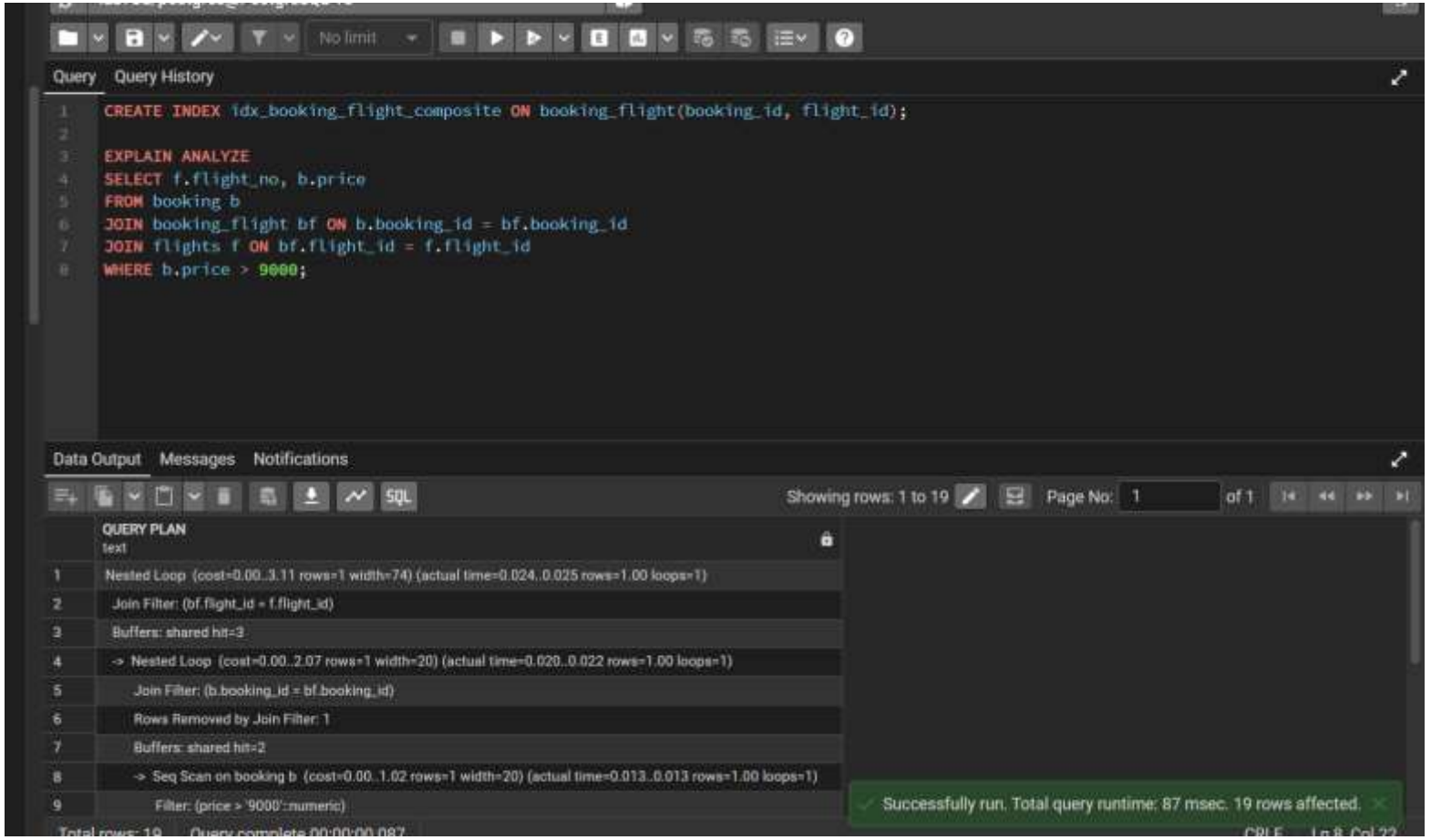
8 task


Query
Query History
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'passengers';
DROP INDEX IF EXISTS idx_passengers_passport;
DROP INDEX IF EXISTS idx_passengers_compound;
Data Output
Messages
Notifications
ЗАМЕЧАНИЕ: индекс "idx_passengers_passport" не существует, пропускается
DROP INDEX
Query returned successfully in 59 msec.
Query returned successfully in 59 msec.

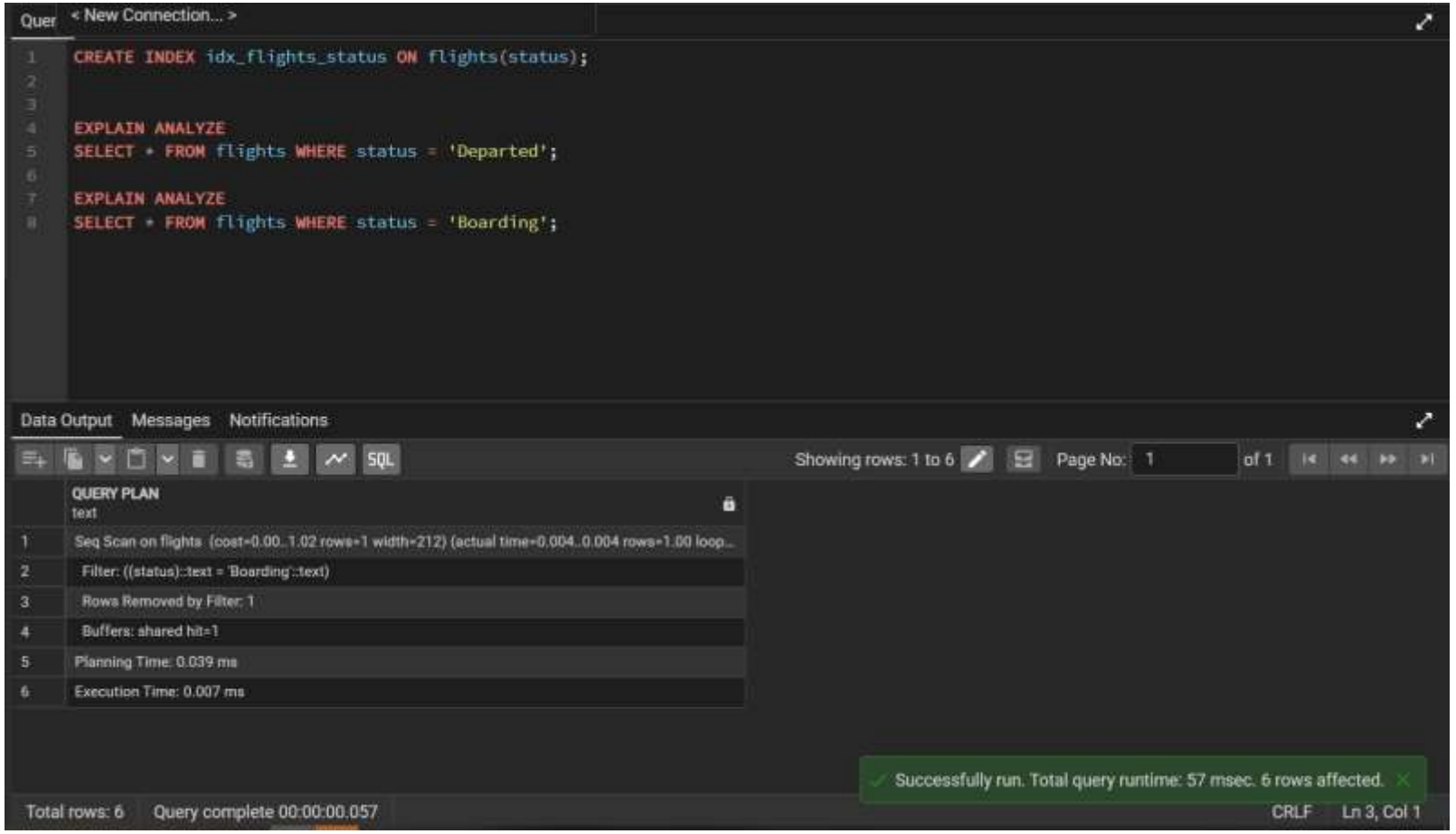
9 task


No limit
Query
Query History
EXPLAIN ANALYZE
SELECT * FROM booking WHERE passenger_id = 100;
CREATE INDEX idx_booking_passenger_id ON booking(passenger_id);
EXPLAIN ANALYZE
SELECT * FROM booking WHERE passenger_id = 100;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 8
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on booking (cost=0.00..1.02 rows=1 width=32) (actual time=0.009..0.010 rows=1.00 loop...
2 Filter: (passenger_id = 100)
3 Rows Removed by Filter: 1
4 Buffers: shared hit=1
5 Planning:
6 Buffers: shared hit=5 read=1
7 Planning Time: 1.671 ms
8 Execution Time: 0.021 ms
Successfully run. Total query runtime: 41 msec. 8 rows affected.
Total rows: 8
Query complete 00:00:00.041
CRLF
Ln 1, Col 1

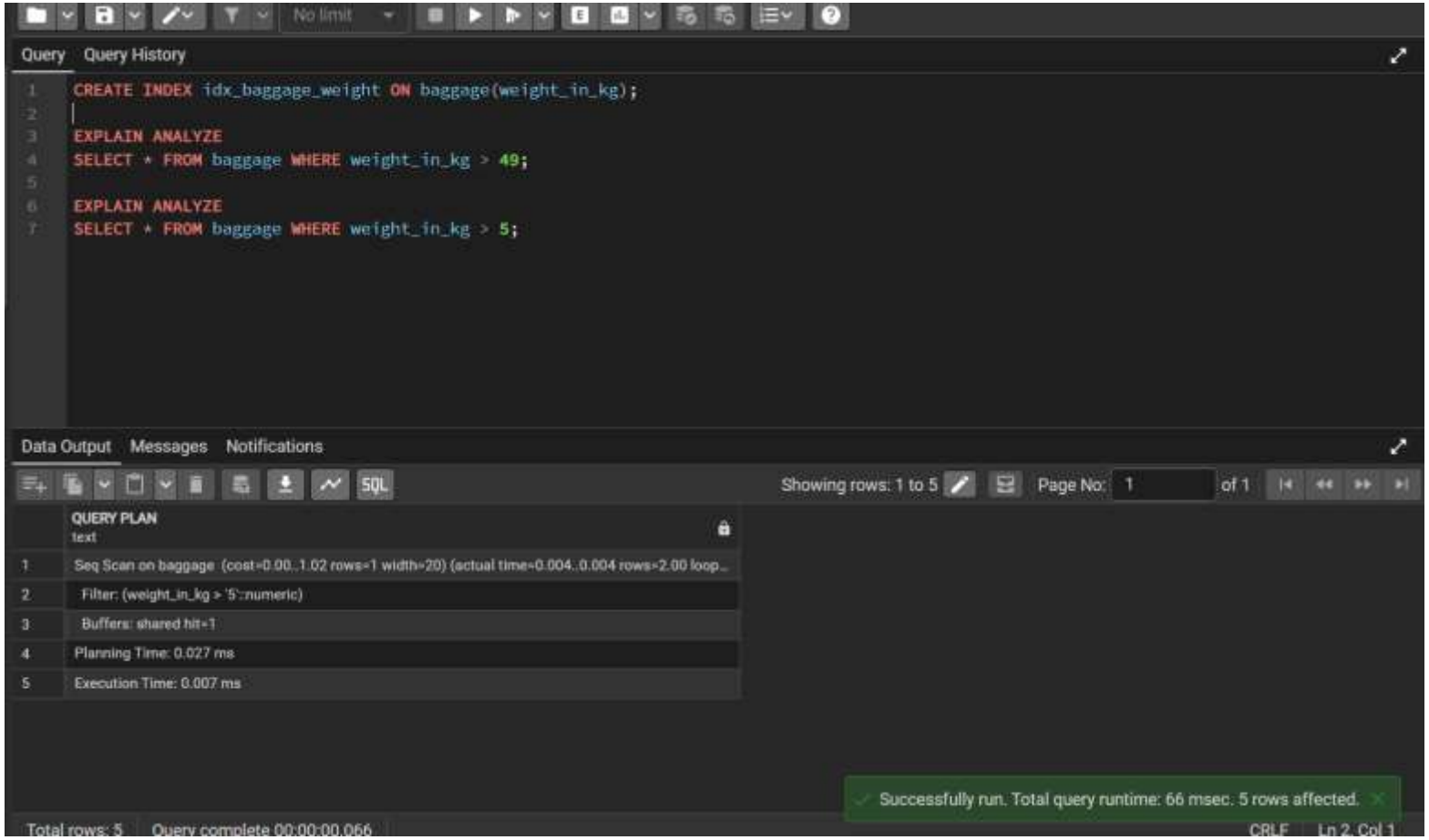
10 task


No limit
Query
Query History
CREATE INDEX idx_booking_flight_composite ON booking_flight(booking_id, flight_id);
EXPLAIN ANALYZE
SELECT f.flight_no, b.price
FROM booking b
JOIN booking_flight bf ON b.booking_id = bf.booking_id
JOIN flights f ON bf.flight_id = f.flight_id
WHERE b.price > 9000;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 19
Page No: 1
of 1
QUERY PLAN
text
1 Nested Loop (cost=0.00..3.11 rows=1 width=74) (actual time=0.024..0.025 rows=1.00 loops=1)
2 Join Filter: (bf.flight_id = f.flight_id)
3 Buffers: shared hit=3
4 -> Nested Loop (cost=0.00..2.07 rows=1 width=20) (actual time=0.020..0.022 rows=1.00 loops=1)
5 Join Filter: (b.booking_id = bf.booking_id)
6 Rows Removed by Join Filter: 1
7 Buffers: shared hit=2
8 -> Seq Scan on booking b (cost=0.00..1.02 rows=1 width=20) (actual time=0.013..0.013 rows=1.00 loops=1)
9 Filter: (price > '9000'::numeric)
Successfully run. Total query runtime: 87 msec. 19 rows affected.

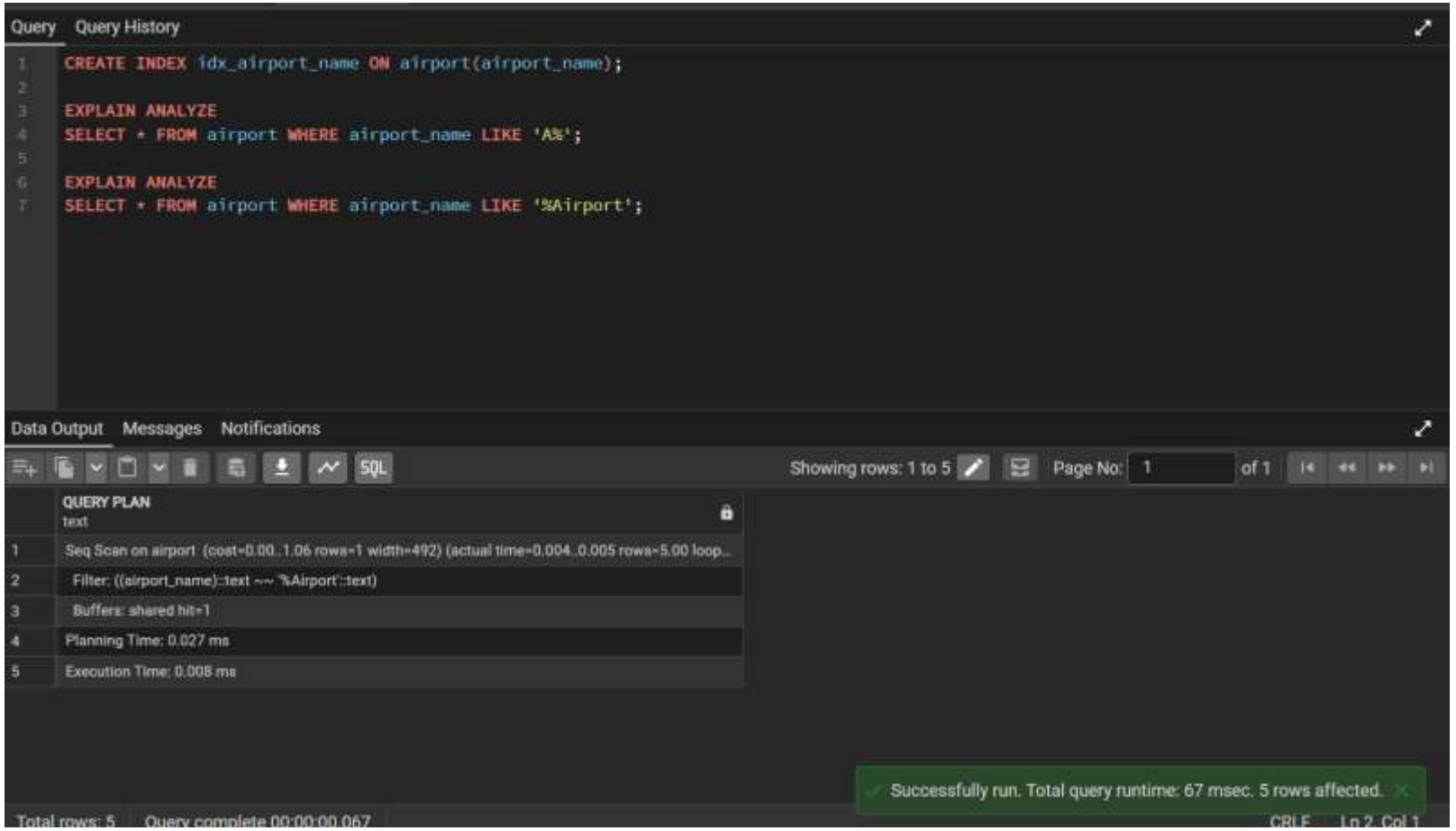
11 task


Quer
< New Connection... >
1 CREATE INDEX idx_flights_status ON flights(status);
2
3
4 EXPLAIN ANALYZE
5 SELECT * FROM flights WHERE status = 'Departed';
6
7 EXPLAIN ANALYZE
8 SELECT * FROM flights WHERE status = 'Boarding';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 6
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on flights (cost=0.00..1.02 rows=1 width=212) (actual time=0.004..0.004 rows=1.00 loop...
2 Filter: ((status)::text = 'Boarding'::text)
3 Rows Removed by Filter: 1
4 Buffers: shared hit=1
5 Planning Time: 0.039 ms
6 Execution Time: 0.007 ms
Successfully run. Total query runtime: 57 msec. 6 rows affected.
Total rows: 6
Query complete 00:00:00.057
CRLF
Ln 3, Col 1

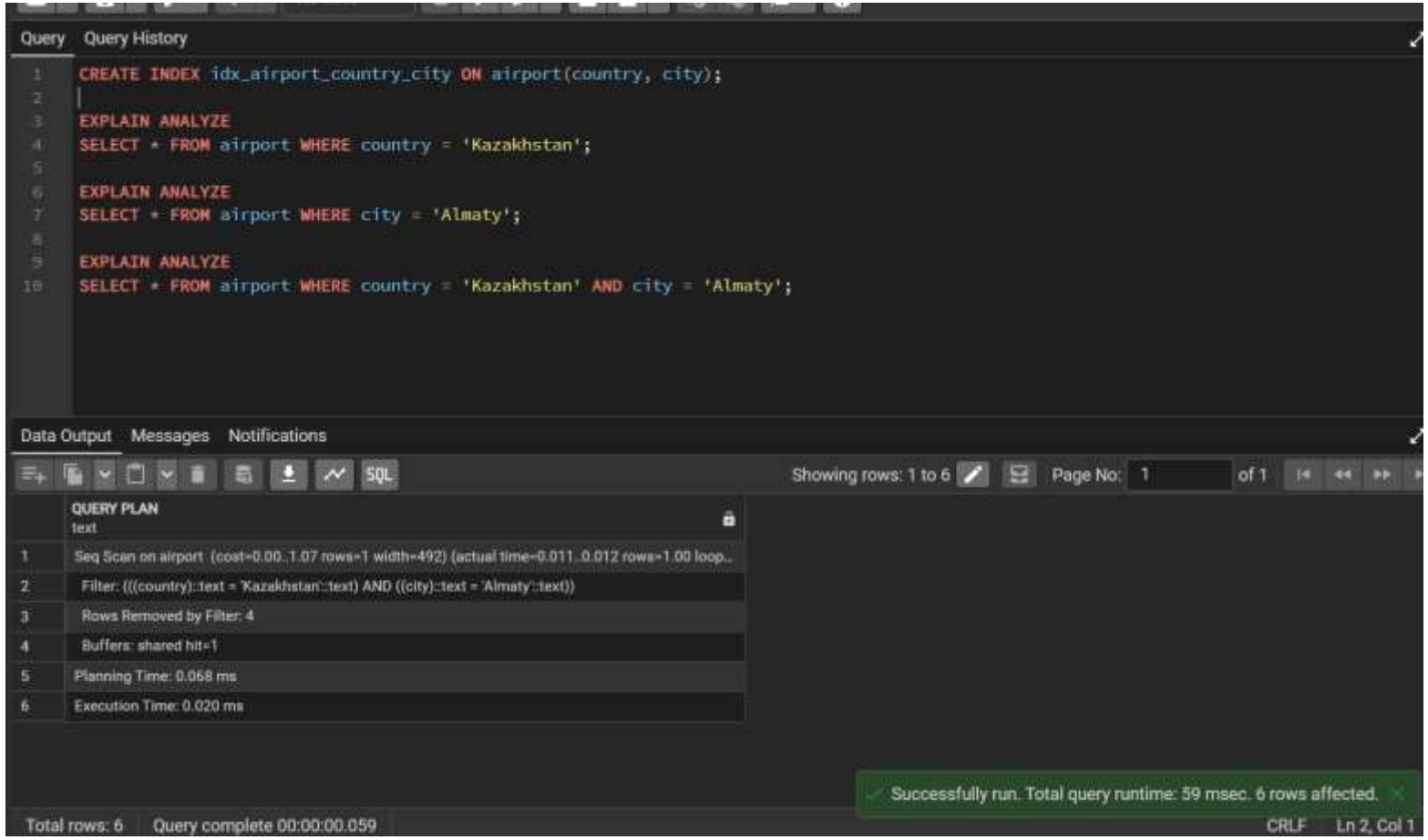
12 task


No limit
Query
Query History
1 CREATE INDEX idx_baggage_weight ON baggage(weight_in_kg);
2
3 EXPLAIN ANALYZE
4 SELECT * FROM baggage WHERE weight_in_kg > 49;
5
6 EXPLAIN ANALYZE
7 SELECT * FROM baggage WHERE weight_in_kg > 5;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 5
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on baggage (cost=0.00..1.02 rows=1 width=20) (actual time=0.004..0.004 rows=2.00 loop...
2 Filter: (weight_in_kg > '5'::numeric)
3 Buffers: shared hit=1
4 Planning Time: 0.027 ms
5 Execution Time: 0.007 ms
Successfully run. Total query runtime: 66 msec. 5 rows affected.
Total rows: 5
Query complete 00:00:00.066
CRLF
Ln 2, Col 1

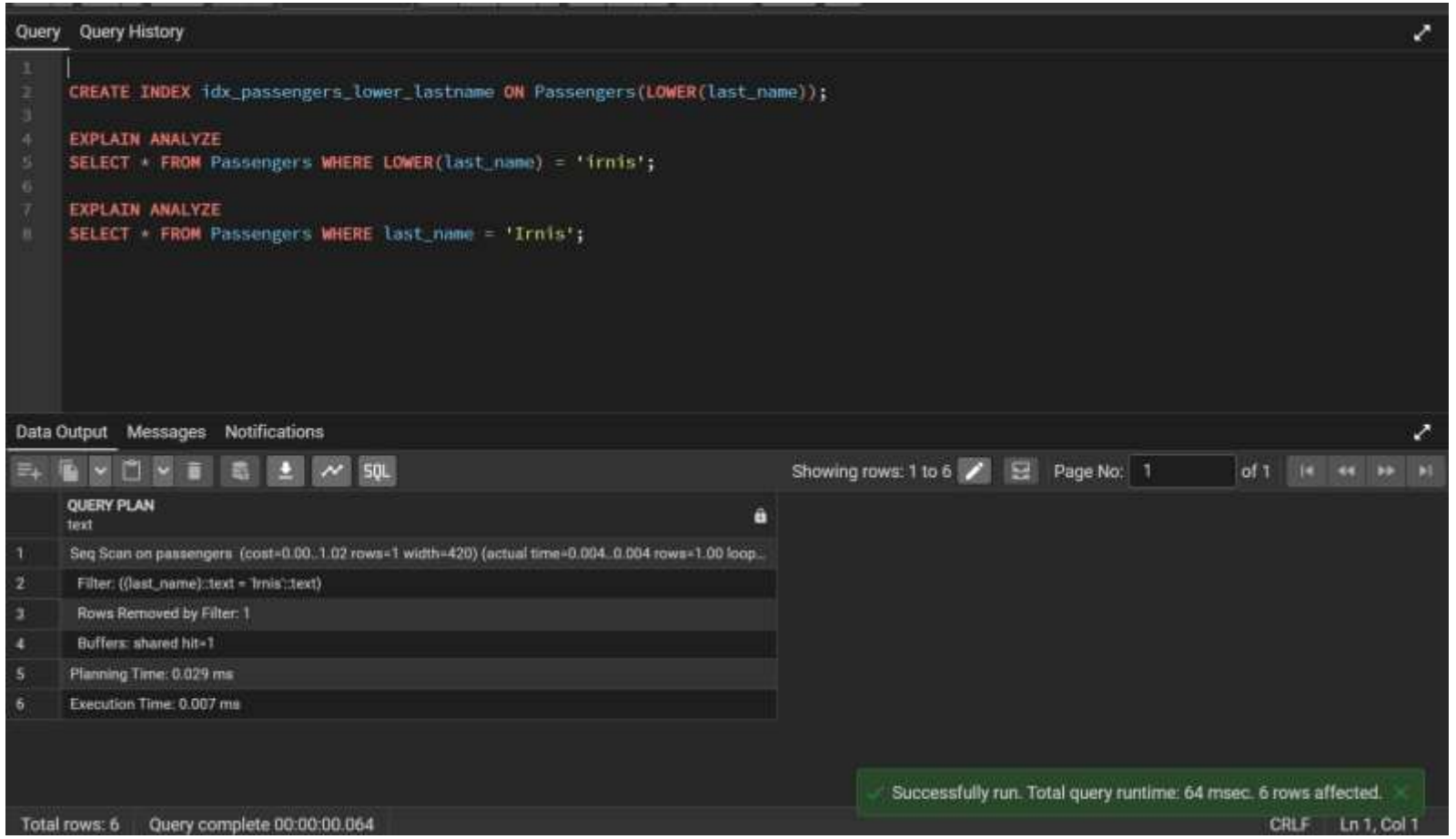
13 task


Query
Query History
CREATE INDEX idx_airport_name ON airport(airport_name);
EXPLAIN ANALYZE
SELECT * FROM airport WHERE airport_name LIKE 'A%';
EXPLAIN ANALYZE
SELECT * FROM airport WHERE airport_name LIKE '%Airport';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 5
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on airport (cost=0.00..1.06 rows=1 width=492) (actual time=0.004..0.005 rows=5.00 loop...
2 Filter: ((airport_name)::text ~~ '%Airport'::text)
3 Buffers: shared hit=1
4 Planning Time: 0.027 ms
5 Execution Time: 0.008 ms
Successfully run. Total query runtime: 67 msec. 5 rows affected.
Total rows: 5
Query complete 00:00:00.067
CRLF
Ln 2, Col 1

14 task


Query
Query History
1 CREATE INDEX idx_airport_country_city ON airport(country, city);
2
3 EXPLAIN ANALYZE
4 SELECT * FROM airport WHERE country = 'Kazakhstan';
5
6 EXPLAIN ANALYZE
7 SELECT * FROM airport WHERE city = 'Almaty';
8
9 EXPLAIN ANALYZE
10 SELECT * FROM airport WHERE country = 'Kazakhstan' AND city = 'Almaty';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 6
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on airport (cost=0.00..1.07 rows=1 width=492) (actual time=0.011..0.012 rows=1.00 loop...
2 Filter: (((country)::text = 'Kazakhstan'::text) AND ((city)::text = 'Almaty'::text))
3 Rows Removed by Filter: 4
4 Buffers: shared hit=1
5 Planning Time: 0.068 ms
6 Execution Time: 0.020 ms
Successfully run. Total query runtime: 59 msec. 6 rows affected.
Total rows: 6
Query complete 00:00:00.059
CRLF
Ln 2, Col 1

15 task


Query
Query History
CREATE INDEX idx_passengers_lower_lastname ON Passengers(LOWER(last_name));
EXPLAIN ANALYZE
SELECT * FROM Passengers WHERE LOWER(last_name) = 'irnis';
EXPLAIN ANALYZE
SELECT * FROM Passengers WHERE last_name = 'Irnis';
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 6
Page No: 1
of 1
QUERY PLAN
text
1 Seq Scan on passengers (cost=0.00..1.02 rows=1 width=420) (actual time=0.004..0.004 rows=1.00 loop...
2 Filter: ((last_name)::text = 'Irnis'::text)
3 Rows Removed by Filter: 1
4 Buffers: shared hit=1
5 Planning Time: 0.029 ms
6 Execution Time: 0.007 ms
Successfully run. Total query runtime: 64 msec. 6 rows affected.
Total rows: 6
Query complete 00:00:00.064
CRLF
Ln 1, Col 1